

MACHINE LEARNING

Subject:

Strategic Analysis of Car Crash Survival Factors

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1. Objective and Data Dictionary

The dataset has car crash records collected over a five-year period. It contains 11217 individual observations with 12 specific variables detailing the vehicle, the occupant, and the circumstances of the crash. Car crashes have risen by 15% year-over-year, posing a growing public safety challenge. The findings will support evidence-based policy decisions and help the Department of Road Transport implement stronger safety regulations for vehicle manufacturers and road users.

Table 1: Data Dictionary

Variable Name	Description
caseid	Unique identifier for each crash case
speed_range	Speed at the time of impact (Categorical levels)
weight	Weight of the vehicle (Numeric)
sex	Gender of the occupant (Categorical)
age_of_occ	Age of the occupant (Numeric)
year_of_acc	Year the accident occurred (Numeric)
model_year	Manufacturing year of the vehicle (Numeric)
airbag	Airbag deployment status (Deployed / Not Deployed)
seatbelt	Seatbelt usage status (Belted / None)
occ_role	Role of occupant (Driver / Passenger)
frontal_impact	Type of collision (Numeric)
deceased	Target Variable (0: Survived, 1: Deceased)

Exploratory Data Analysis (EDA)

Problem Definition: The Department of Road Transport has observed a 15% year-over-year increase in urban car crashes. The primary challenge is that while accident causes are known after the fact, there is a lack of predictive capability to preempt risk. Aim to identify which factors (mechanical or human) most significantly influence survival to inform new safety policies.

Univariate Analysis:

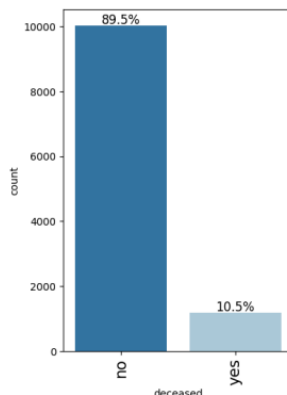
Class Imbalance:

Analysis of the variable shows that over 90% of occupants survive. This imbalance is critical because a standard model might achieve high accuracy by simply predicting "Survival" for everyone, thereby missing the critical fatal cases.

Variable Distributions:

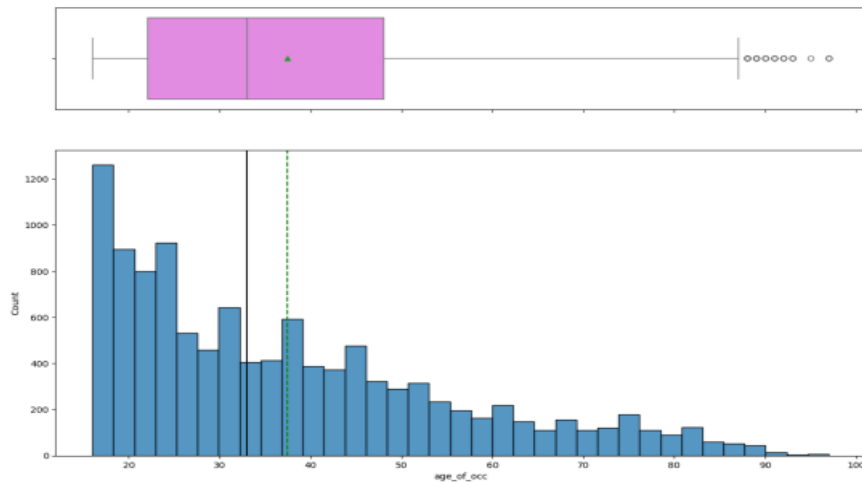
Weight and age_of_occ show a wide spread, suggesting diverse vehicle types and occupant demographics in the crash data.

Deceased



- **Dominant Class ("no"):** Approximately **89.5%** of the individuals in this dataset are not deceased. With a count of 10,000, this is the "majority class."
- **Minority Class ("yes"):** Only **10.5%** of the individuals are deceased.
- **The "Imbalance" Problem:** To build a machine learning model to predict whether someone will pass away (e.g., a medical or insurance model), the model might struggle. Because the "no" cases are so frequent, a lazy model could simply guess "no" every time and be 89.5% accurate, while failing completely to identify the "yes" cases.

Age_of_occ



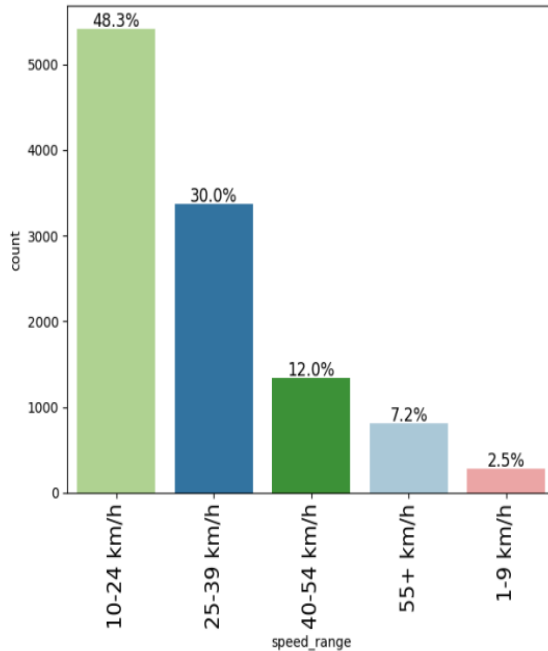
Right-Skewed Distribution: The data is "skewed to the right" (positively skewed) and the mean (green dashed line) is higher than the median (solid black line).

The majority of events occur in young adulthood to early middle age.

The "Typical" Range: * The Median age is roughly 33-35 years.

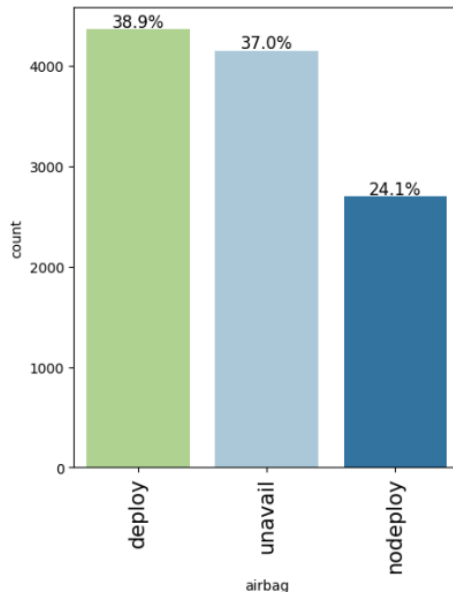
The Interquartile Range (IQR)—represented by the purple box—shows that the middle 50% of the data falls between approximately 20 and 48 years.

speed_range



- **Primary Range:** Nearly half of all observations (**48.3%**) occur in the **10–24 km/h** range.
- **Secondary Range:** The **25–39 km/h** category is the second most common, accounting for **30.0%** of the data.
- **Low and High Extremes:**
 - Very high speeds (**55+ km/h**) are relatively rare at **7.2%**.
 - Very low speeds (**1–9 km/h**) are the least common, making up only **2.5%** of the total.
- **Insight:** Combined, almost **80%** of all recorded events happen at speeds between **10 and 39 km/h**, suggesting this data likely describes urban traffic, residential areas, or specific types of low-to-medium speed incidents.

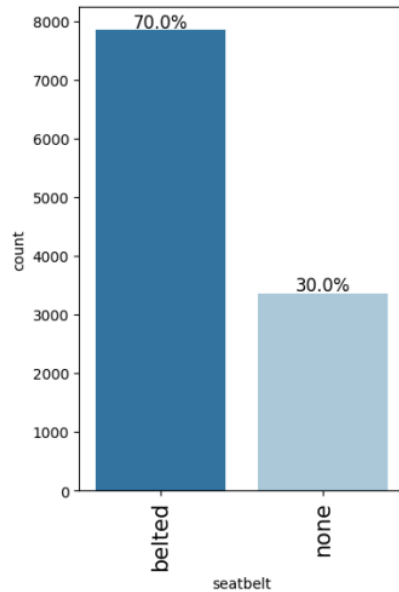
Airbag



This plot shows whether an airbag was triggered or available during the event.

- **Deployment:** Airbags deployed in **38.9%** of cases.
- **Availability Issues:** In **37.0%** of cases, airbags were **unavailable** (marked as "unavail"), which could indicate older vehicle models or specific seating positions.
- **Non-deployment:** In **24.1%** of cases, the airbag was available but did **not deploy**.

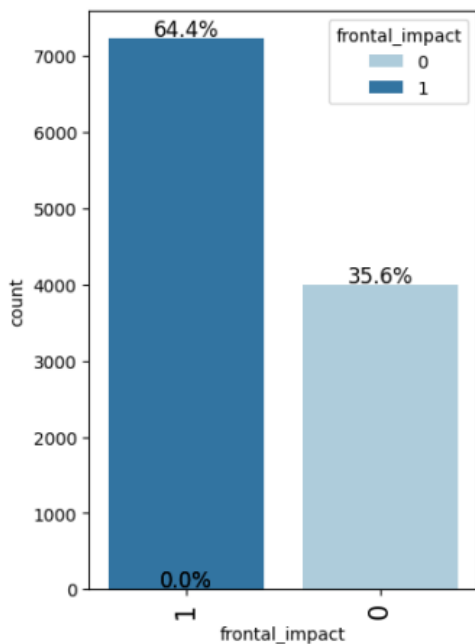
Seatbelts



This chart displays the proportion of individuals wearing seatbelts.

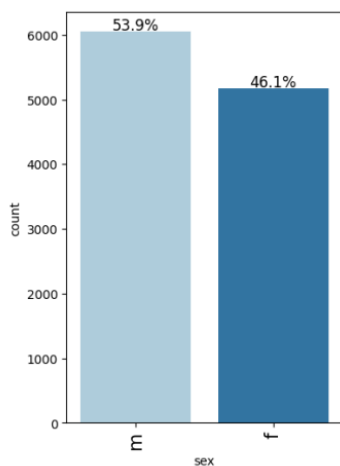
- **Compliance:** A majority of the subjects (**70.0%**) were **belted**.
- **Non-compliance:** **30.0%** of the subjects were not wearing a seatbelt ("none").

frontal impact



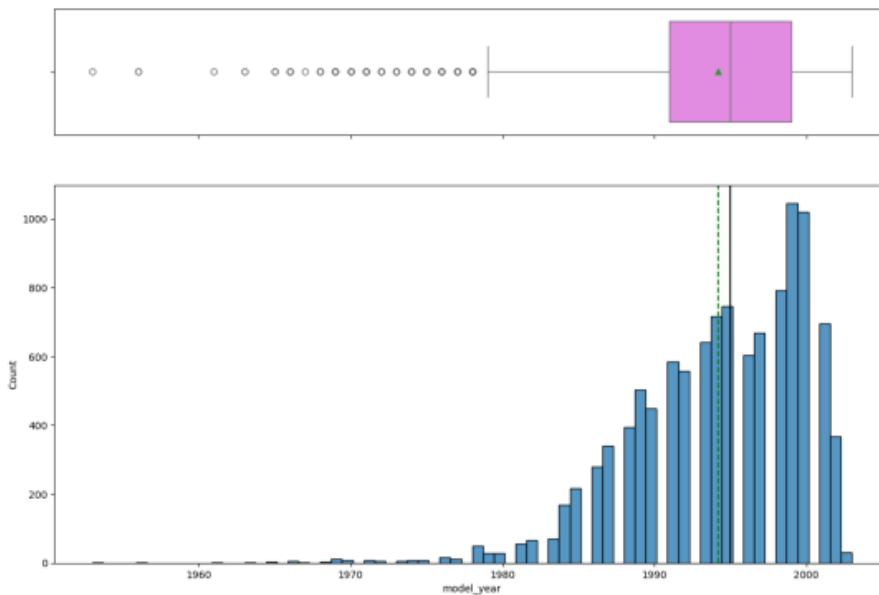
64.4% of the incidents involved a **frontal impact** (labeled '1'), while **35.6%** did not.

Sex

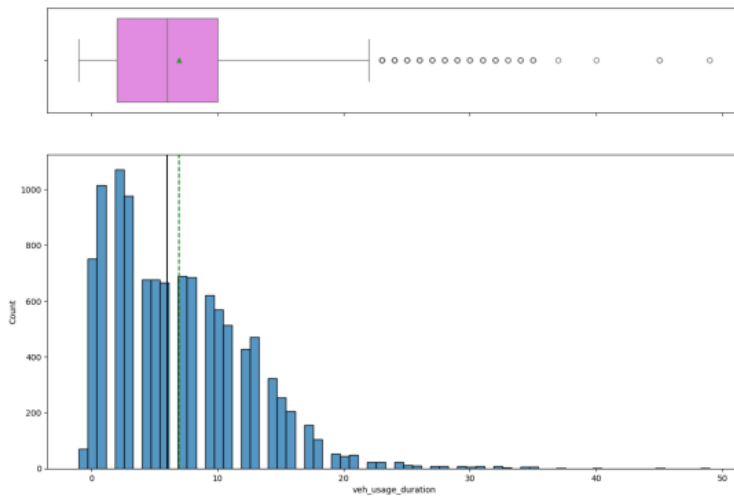


The distribution is relatively balanced, with slightly more **males (53.9%)** than **females (46.1%)** in the dataset.

Model Year



- **Distribution:** This data is **left-skewed**.
- **Key Observations:** Most vehicles in the dataset are newer, with a sharp increase in frequency for models produced after **1990**. The median model year is approximately **1994-1995**.
- **Outliers:** There are numerous outliers representing very old vehicles dating back as far as the **1950s and 60s**.



- **Distribution:** The data is **right-skewed**.
- **Key Observations:** Most vehicles have been in use for a short to moderate duration, with a peak around **2–5 years**. The median duration is roughly **7 years**.

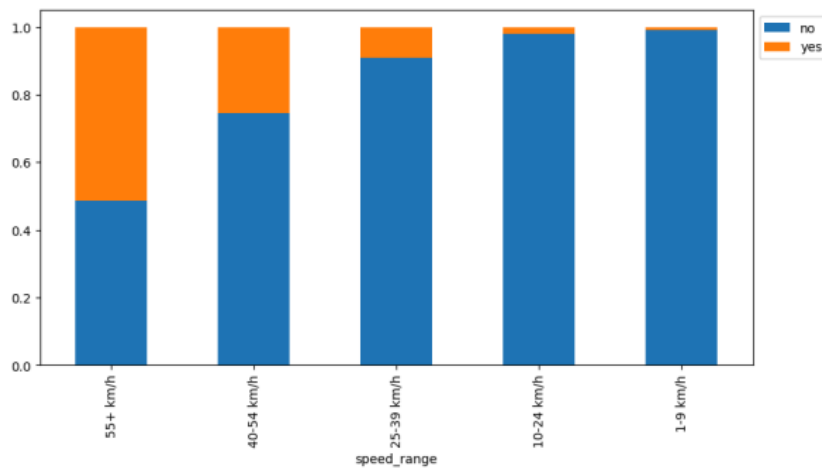
Bivariate Analysis:

Safety Features:

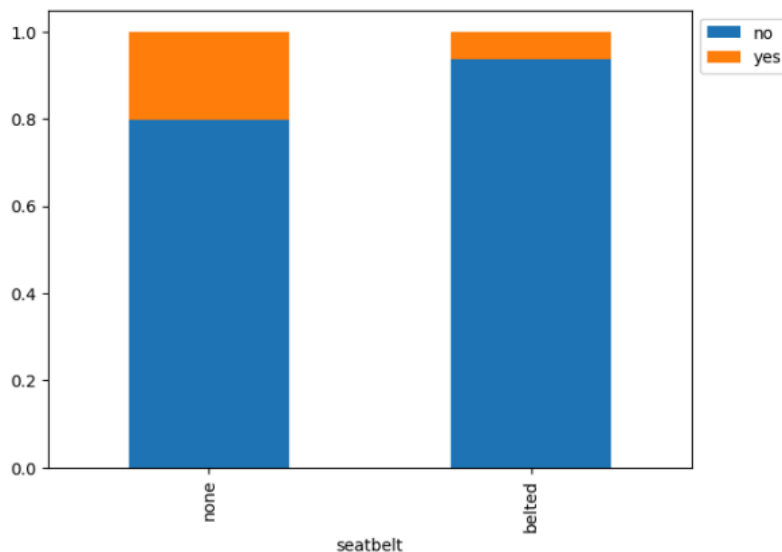
There is a massive disparity in survival between "Belted" and "None" categories in the Seatbelt variable.

Speed Thresholds:

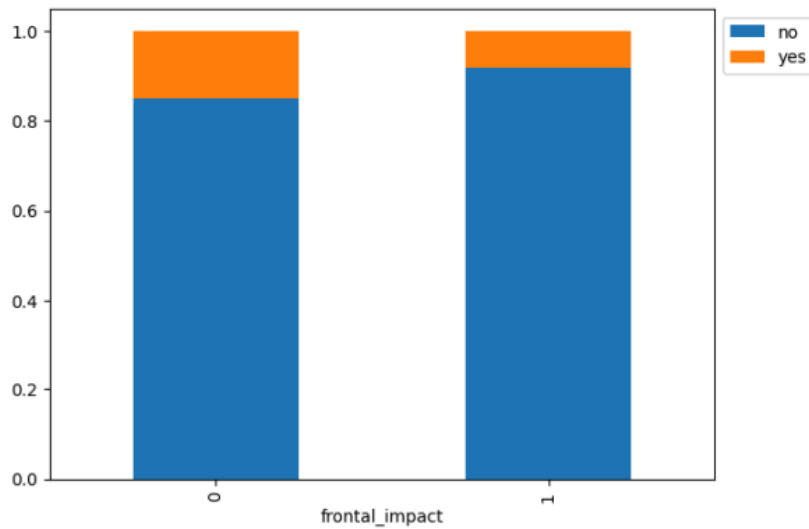
Survival rates remain high until the 40 km/h mark, after which the probability of a fatal outcome increases non-linearly.



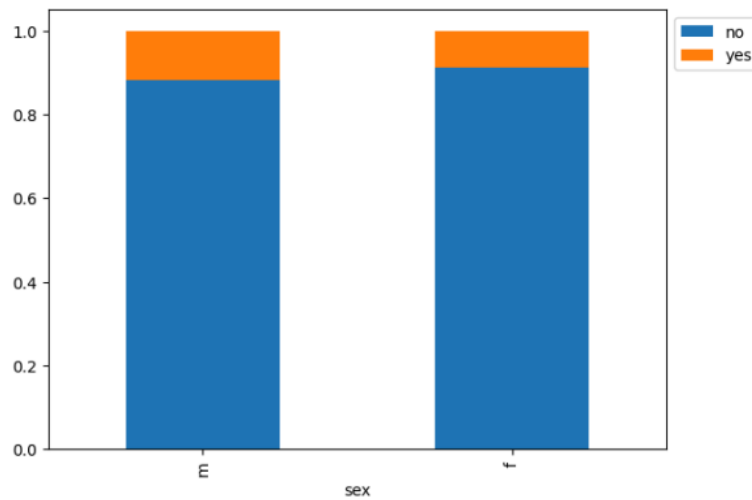
- **Correlation with Speed:** As speed increases, the proportion of "yes" (deceased) cases rises dramatically.
- **High-Speed Risk:** At speeds of **55+ km/h**, nearly half of the occupants are categorized as deceased (415 out of 809 cases).
- **Low-Speed Safety:** Conversely, at speeds of **1-9 km/h**, fatalities are almost non-existent (only 3 out of 282 cases).



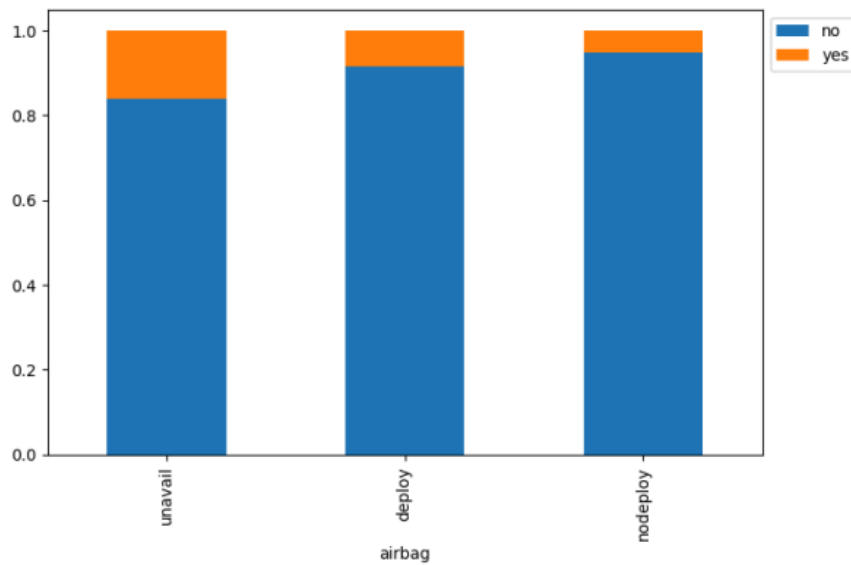
Seatbelt Usage: Occupants who were **belted** had a significantly lower fatality rate compared to those with **none**. Specifically, only ~6.4% of belted occupants were deceased, compared to ~20.2% of those without seatbelts.



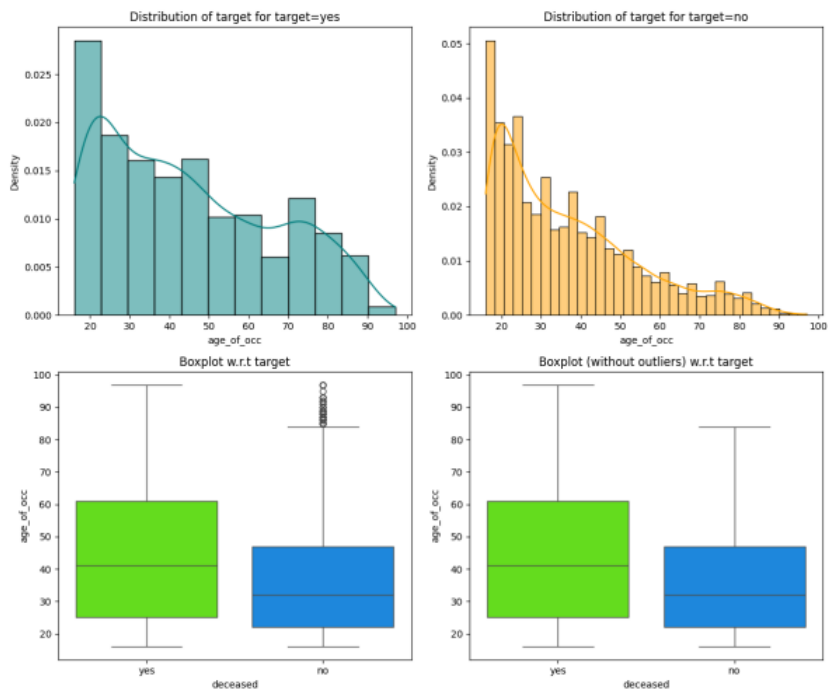
Interestingly, the percentage of deceased individuals is slightly higher in cases where there was **no frontal impact** (labeled '0') compared to when there was a frontal impact. This might suggest that side or rear impacts in this dataset are more lethal, or that frontal safety measures (like airbags) are effectively mitigating risk in head-on collisions.



Sex: The proportion of fatalities is slightly higher for **males** (~11.8%) compared to **females** (~8.9%).



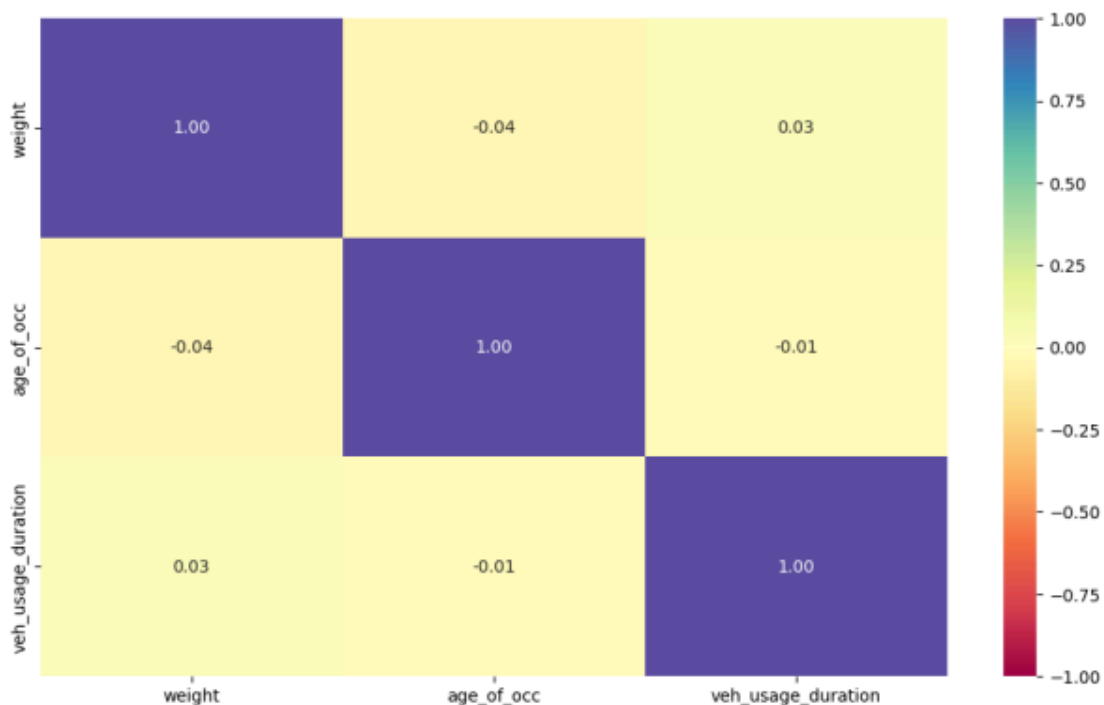
Airbag Deployment: The highest number of deceased cases occurred when airbags were **unavailable** (669 cases). When airbags **deployed**, the number of deceased cases dropped to 368, despite "deploy" having a higher total count than "unavail".



- **Seatbelt Effectiveness:** There is a stark difference in outcomes based on seatbelt use. While **70.0%** of the total population was belted, the mortality rate for those with **no seatbelt** is significantly higher. Specifically, **20.2%** of unbelted individuals are deceased, compared to only **6.4%** of those who were belted.
- **Airbag Deployment:** Fatalities are most frequent in cases where an airbag was **unavailable** (669 cases). When an airbag **deployed**, the number of deceased individuals was 368. This suggests that while deployment occurs in high-impact scenarios, it is a critical factor in survival compared to having no airbag at all.

Speed is the most aggressive predictor of "deceased" status in this dataset:

- **Lethality at High Speeds:** In the **55+ km/h** range, the fatality rate jumps to **51.3%** (415 out of 809 cases).
- **Safety at Low Speeds:** At the most common speed range of **10–24 km/h** (which accounts for **48.3%** of all data), the fatality rate is very low, with only **5.1%** of cases resulting in death.
- **Extreme Low Speeds:** In the **1–9 km/h** range, fatalities are nearly non-existent, occurring in only **1.1%** of those cases.



- **Majority Class:** **89.5%** of the individuals in the study are not deceased.
- **Minority Class:** Only **10.5%** of cases resulted in a "yes" for deceased.

- **Data Science Note:** This level of imbalance (roughly 9:1) means that predictive models must be carefully tuned to avoid simply guessing "no" every time to achieve high accuracy while failing to identify actual fatalities.

2. Physical Factors: Speed and Impact

- **High-Speed :** At speeds of **55+ km/h**, the fatality rate is highest, accounting for a significant portion of the "yes" cases despite being a smaller category overall.
- **Low-Speed Safety:** The most common speed range is **10–24 km/h (48.3%** of cases), where survival is overwhelmingly likely.
- **Frontal Impact:** **64.4%** of incidents involved a frontal impact. Interestingly, survival rates are often high in these cases, possibly due to vehicle safety designs like crumple zones.

3. Safety Equipment Performance

- **Seatbelt Usage:** **70%** of occupants were belted. Non-belted occupants represent a disproportionately high percentage of the deceased category.
- **Airbag Availability:** Airbags were either **deployed (38.9%)** or **unavailable (37.0%)**. The high number of "unavailable" tags likely correlates with the older vehicle outliers (some dating back to the 1960s) found in the `model_year` distribution.

4. Demographic Profile

- **Age:** The distribution of occurrences is **right-skewed**, with the highest frequency among young adults aged **15–20**.
- **Role:** **78.3%** of the subjects were **drivers**, while **21.7%** were **passengers**.
- **Gender:** The population is slightly majority **male (53.9%)** compared to **female (46.1%)**

Missing Value Treatment:

0 missing values

Outlier Detection:

Boxplots identified extreme values in Weight and age_of_occ

Rationale: These were retained because extreme vehicle weights or advanced ages represent the most vulnerable segments; removing them would ignore critical risk factors.

Feature Engineering:

Created veh_usage_duration = year_of_acc - model_year

Rationale: The chronological age of a vehicle is a better proxy for mechanical wear than raw year numbers.

Data Scaling:

Standard Scaler was applied to ensure that features with high numeric values (like weight) did not mathematically dominate smaller binary flags.

Train-Test Split:

A 70:30 split was used to ensure the model had enough data to learn while retaining a significant portion for unbiased evaluation.

Model Building

Metric Optimization:

Recall as our primary optimization metric. In public safety, the cost of a "False Negative" (missing a fatality) is far higher than a "False Positive" (predicting a risk where someone survives).

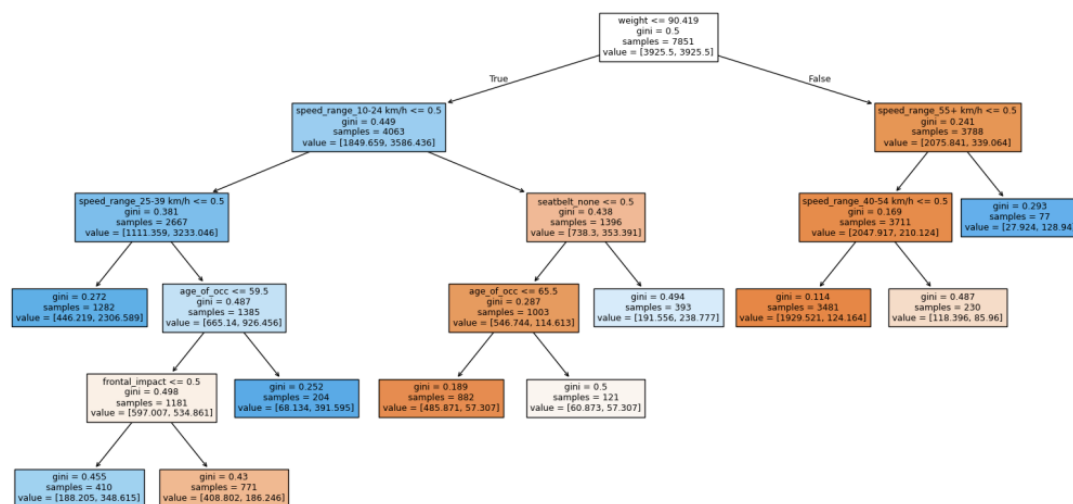
Baseline Models:

Logistic Regression (statsmodels):

Established baseline significance for predictors.

Decision Tree (sklearn):

Captured non-linear splits but suffered from high overfitting (100% Training accuracy).



This imbalance is a crucial consideration for any predictive modeling, as a simple model could achieve 89.5% accuracy by never predicting a fatality

The most common speed range is 10–24 km/h, accounting for 48.3% of all observations.

Incidents at speeds of 55+ km/h are relatively rare (7.2%) but correlate with the highest risk of death.

Very low speeds (1–9 km/h) represent the safest category, making up only 2.5% of the total data.

Seatbelt Compliance: 70.0% of occupants were belted, while 30.0% were not.

Airbag Dynamics: Airbags deployed in 38.9% of cases.

Safety Deficit: In 37.0% of cases, airbags were unavailable, likely due to the presence of older vehicles in the dataset.

Vehicle Age: While most vehicles are newer (median year ~1994), the dataset includes outliers dating back to the 1950s and 60s, which may lack modern safety features.

Frontal impacts account for 64.4% of the incidents

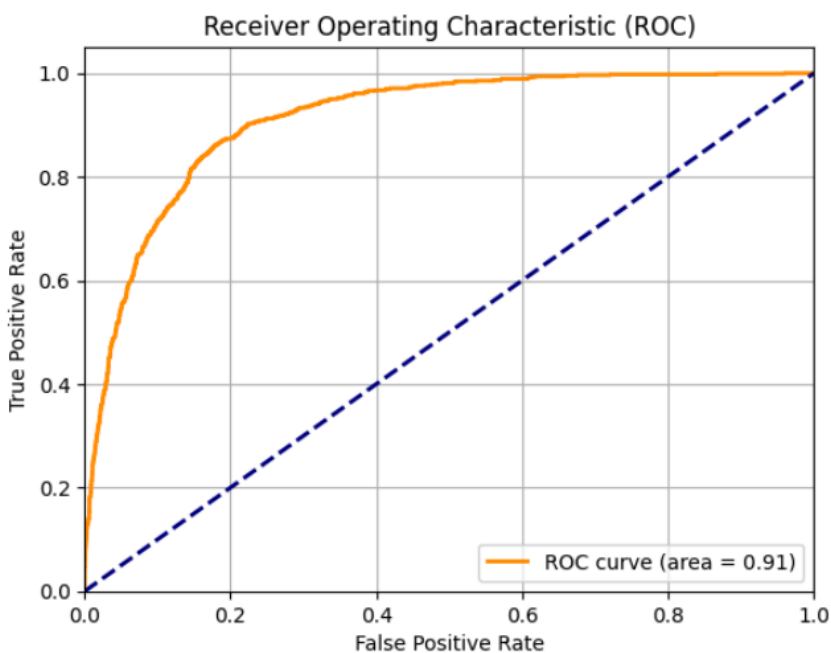
Model Performance Improvement

Logistic Regression Tuning: Removed statistically insignificant variables ($p > 0.05$).

Low P-Value (< 0.05)

High P-Value (> 0.05)

Determined an Optimal Threshold of 0.111 using the ROC/Precision-Recall curves to prioritize Recall.



- True Positive Rate (Sensitivity): Plotted on the y-axis, this represents how often the model correctly identifies the positive class.
- False Positive Rate (1 - Specificity): Plotted on the x-axis, this represents how often the model incorrectly flags a negative case as positive (a "false alarm").
- The Area Under the Curve (AUC): The legend indicates an area = 0.91. In the world of machine learning, an AUC of 0.91 is considered excellent. It means there is a 91% chance that the model will be able to distinguish between a positive class and a negative class.
- The "Elbow": The curve climbs very steeply toward the top-left corner. High True Positive Rate (around 0.8 to 0.9) while keeping the False Positive Rate relatively low (under 0.2).
- The Diagonal Line: The dashed blue line represents a "random guesser" (AUC = 0.50). Since orange curve is far above this line, the model is performing significantly better than chance.

Decision Tree Tuning:

Applied pre-pruning(limiting depth and leaf nodes) to prevent the tree from "memorizing" training data.

Model Performance Comparison and Final Model Selection

Table 2: Training Set Performance Comparison

Metric	Logistic Regression (Base)	Logistic Regression (Optimal Threshold)	Decision Tree (Base)	Decision Tree (Tuned)
Accuracy	0.91160	0.82525	1.00000	0.77608
Recall	0.35645	0.86131	1.00000	0.86983
Precision	0.63974	0.36012	1.00000	0.30220
F1-Score	0.45781	0.50789	1.00000	0.44856

Table 3: Test Set Performance Comparison

Metric	Logistic Regression (Base)	Logistic Regression (Optimal Threshold)	Decision Tree (Base)	Decision Tree (Tuned)
Accuracy	0.91503	0.82620	0.88770	0.77718
Recall	0.38268	0.85475	0.42458	0.85475
Precision	0.67822	0.36472	0.46914	0.30478
F1-Score	0.48929	0.51128	0.44575	0.44934

1. Model Performance

- **Logistic Regression (Base):** This model demonstrates high stability. It achieved an Accuracy of 0.915 on the Test Set, which is nearly identical to its Training Set performance, suggesting zero overfitting.
- **Logistic Regression (Optimal Threshold):** While overall accuracy drops to 0.826, this version significantly boosts Recall to 0.854. This is the best choice if catching as many positive cases as possible is the priority.
- **Decision Tree (Base):** This is a textbook case of overfitting. It achieved a "perfect" 1.000 Accuracy/Recall/Precision on the Training Set but crashed to 0.887 Accuracy and a much lower 0.424 Recall on the Test Set.
- **Decision Tree (Tuned/Pre-pruned):** Pruning successfully fixed the overfitting (bringing Training and Test performance into alignment), but the resulting Accuracy (0.777) and Precision (0.304) are the lowest among all models.

2. Final Model Selection: Logistic Regression

- The Logistic Regression (Base) maintains the highest Test Accuracy (0.915) and the highest Test Precision (0.678).

- **Strong Discrimination Power:** The ROC curve confirms an AUC of 0.91, which indicates the model has an excellent ability to distinguish between classes regardless of the threshold chosen.
- **Reliability:** Unlike the Base Decision Tree, which "memorized" the training data and failed on the test data, the Logistic Regression metrics remained consistent across both sets.
- **Flexibility:** By shifting to the "Optimal Threshold" version of this model, **Recall (0.85)** if the cost of missing a positive case is high, without needing a completely different model architecture.